On Hamilton-Jacobi equations for the N-body problem

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N-*body problem*

We consider the classical Newtonian N-body problem in $\mathbb{R}^d$, with $d \geq 2$, where N point masses $m_i > 0$ occupy positions $r_i \in \mathbb{R}^d$:

$$m_i \ddot{r}_i = -\sum_{j \neq i} m_i m_j \frac{r_i - r_j}{|r_i - r_j|^3}, \quad \text{for every } \; i = 1, \dots, N. \tag{Eq.}$$

In more compact notation:

$$\mathcal{M} \ddot{x} = \nabla U(x), \quad \text{with} \quad U(x) = \sum_{1 \leq i < j \leq N} \frac{m_i m_j}{|r_i - r_j|},$$

$x = (r_1, \dots, r_N) \in \mathbb{R}^{dN}$ and $\mathcal{M} = \operatorname{diag}(m_1 I_d, \dots, m_N I_d)$.

- Denote by $\mathcal{X}$ the $d(N-1)$-dimensional space of configurations with center of mass fixed at the origin:

$$x \in \mathcal{X} \iff \sum_{i=1}^{N} m_i r_i = 0.$$

- In $\mathcal{X}$, we define Ω, the set of collison-free configurations:

$$x \in \Omega \iff r_i \neq r_j \quad \text{for all} \quad i \neq j.$$

- By L and H, we indicate the **Lagrangian** and **Hamiltonian** of the system:

$$L(x, q) = \frac{\|q\|^2_{\mathcal{M}}}{2} + U(x) \quad \text{and} \quad H(x, p) = \frac{\|p\|^2_{\mathcal{M}^{-1}}}{2} - U(x).$$

Expansive motions for (Eq.)

A motion $\gamma : [t_0, +\infty) \to \Omega$ is **expansive** if all mutual distances diverge:

$$|r_i(t) - r_j(t)| \to \infty, \text{ as } \; t \to \infty, \quad \text{for all} \quad i \neq j.$$

Conservation of energy $H(\gamma(t), \mathcal{M}\dot{\gamma}(t))$ implies that expansive motions **can only exist at nonnegative energy levels**.

Results on expansive motions:

Asymptotic behavior [Po, MS]

- $\gamma(t) = a\,t + O(t^{2/3})$, as $t \to \infty$, for some $a \in \mathcal{X}$.

Chazy's classification [Ch]:

- **Hyperbolic** (H): $a \in \Omega$ and $|r_i - r_j| \approx t$;
- **Completely parabolic** (P): $a = 0$ and $|r_i - r_j| \approx t^{2/3}$;
- **Hyperbolic-parabolic** (HP): $a \in \mathcal{X} \setminus \Omega$, $a \neq 0$.

From the classification, it follows that

$$H(\gamma(t), \mathcal{M}\dot{\gamma}(t)) = \frac{\|a\|^2_{\mathcal{M}}}{2} =: h \geq 0. \tag{1}$$

Existence. Maderna-Venturelli proved existence for (H) and (P) (see [MV20, MV09]). Polimeni-Terracini via a **unifed variational framework**, improved sharp asymptotics for (P), and gave existence for general case (HP) (see [PT]).

Variational Setting

Given a reference path r_0 (accordingly to the Chazy's classification), and an initial configuration x, the main strategy is to consider solutions $\gamma : [1, \infty) \to \mathcal{X}$, of the form:

$$\gamma_\varphi(t) = r_0(t) + \varphi(t) + x - r_0(1),$$

where:

- $\varphi \in D^{1,2}_0(1, \infty)$:= the subset of $H^1_{\mathrm{loc}}(1, \infty)$ defined by

$$\varphi(1) = 0, \quad \|\varphi\|^2_{\mathcal{D}} := \int_1^\infty \|\dot{\varphi}(t)\|^2 \, dt < \infty.$$

On $\mathcal{D}^{1,2}_0(1, \infty)$, it is well-defined the **renormalized functional**

$$\mathcal{A}^{\mathrm{ren}}_x(\varphi) := \int_1^\infty \Big[\frac{\|\dot{\varphi}(t)\|^2_{\mathcal{M}}}{2} + U(\gamma_\varphi(t)) - U(r_0(t)) - \langle \ddot{r}_0(t), \varphi(t)\rangle_{\mathcal{M}}\Big] \, dt.$$

Renormalized Action Principle [PT]

If $\varphi \in \mathcal{D}^{1,2}_0(1, \infty)$:

$$\mathcal{A}^{\mathrm{ren}}_x(\varphi) = \min_{\psi \in D^{1,2}_0(1,\infty)} \mathcal{A}^{\mathrm{ren}}_x(\psi),$$

then γ_φ is a *free-time* minimizer of the Lagrangian action at energy h, i.e. for all $[a, b] \in (1, \infty)$, γ_φ minimizes

$$\eta \mapsto \int_{I_\eta} L(\eta(t), \dot{\eta}(t)) \, dt + h|I_\eta|,$$

among all $\eta \in AC(I_\eta)$, with end-points $(\gamma_\varphi(a), \gamma_\varphi(b))$; γ_φ satisfies (Eq.) in $(1, \infty)$.

$\mathcal{A}^{\mathrm{ren}}_x$ is proved to be **coercive** for all $x \in \mathcal{X}$, and it is of **class** C^1 for any $x \in \Omega$ [PT]; in particular, it is well-defined the function $v : \Omega \to \mathbb{R}$:

$$v(x) = \min_{\varphi \in \mathcal{D}^{1,2}_0(1,\infty)} \mathcal{A}^{\mathrm{ren}}_x(\varphi) - \langle a, x\rangle_{\mathcal{M}}. \tag{2}$$

We are interested in the regularity of v, as solution of Hamilton-Jacobi equation for the N-body problem.

A main ingredient, when working in $D^{1,2}_0(1, \infty)$, is the **Hardy inequality** [BDFT]:

$$\int_1^\infty \frac{\|\varphi(t)\|^2_{\mathcal{M}}}{t^2} \, dt \leq 4 \int_1^\infty \|\dot{\varphi}(t)\|^2_{\mathcal{M}} \, dt.$$

The main result

- Σ = **irregular points**: $x \in \Omega$ in which $\mathcal{A}^{\mathrm{ren}}_x$ admits multiple minimizers.
- Γ = **conjugate points**: $x \in \Omega$ such that:
  - if φ^x is a mimizer of $\mathcal{A}^{\mathrm{ren}}_x$, then $d^2\mathcal{A}^{\mathrm{ren}}_x(\varphi^x)$ is not invertible;
  - the set of minimizers of $\mathcal{A}^{\mathrm{ren}}_x$ is isolated on the set of critical points.

Theorem 1 (B.–Polimeni–Terracini, 2025)

In all cases (H), (P) and (HP), the function $v : \Omega \to \mathbb{R}$ defined in (2) is a viscosity solution of the Hamilton-Jacobi equation

$$H(x, \nabla v(x)) = h, \tag{HJ}$$

where h is given in (1).

Moreover:

- *The singular set $\Sigma \cup \Gamma$ is countably $\mathcal{H}^{d(N-1)-1}$-rectifiable;*
- $\dim_{\mathcal{H}}(\Gamma \setminus \Sigma) \leq d(N-1) - 2$.

Comments on Theorem 1. Theorem 1 in [BPT] extends to the N-body problem with singular potentials and non-compact time-infinite expansive motions the results of Cannarsa and Sinestrari [CS], originally established for the evolutionary equations of the type $\partial_t u + H(x, \nabla u) = 0$ in the context of **finite-time non-singular** actions with **time-dependent endpoints**.

Outline of the proof of Theorem 1

- **The value function (2) is a viscosity solution of (HJ).**

The value function $v(x)$ is obtained as the limit, as $T \to \infty$, of **finite-horizon** value functions $\tilde{v}(x, T)$, associated with **truncated renormalized action functionals** enjoying **uniform coercivity estimates** and a decomposition into a positive definite quadratic form plus a controlled term.

Each $\tilde{v}(x, T)$ satisfies a evolutionary Hamilton–Jacobi equation with a remainder converging uniformly in x to $\frac{\|a\|^2_{\mathcal{M}}}{2}$.

By combining **stability of viscosity solutions** with the uniform convergence on compact sets, we conclude that v is a viscosity solution of (HJ).

- **Regularity of the value function.**

Rectifiability of $\Sigma \setminus \Gamma$. Let (x, φ^x) be such that $x \in \Sigma \setminus \Gamma$ and φ^x one of its corresponding minimizers. Applying the **Implicit Function Theorem** to the level 0 of

$$(y, \eta) \mapsto d\mathcal{A}^{\mathrm{ren}}_y[\eta]$$

we get neighborhoods $\mathcal{U}_x$ of x and $\mathcal{U}_{\varphi^x} \subset \mathcal{D}^{1,2}_0(1, \infty)$ of φ^x, such that

$$\mathcal{U}_x \ni y \mapsto \eta = \eta(y) \quad \text{is the unique minimizer in } \mathcal{U}_{\varphi^x},$$

with $\eta(x) = \varphi^x$, and η smooth. **Compactness** of $d\mathcal{A}^{\mathrm{ren}}_x$ implies that

$$v(x) = \min_{i=1,\dots,k} v_i(x), \quad \text{where} \quad v_j(y) = \min_{\varphi \in \mathcal{U}_{\varphi_j}} \mathcal{A}^{\mathrm{ren}}_y(\varphi),$$

for $v_j \in C^\infty$ (since the minimizer is unique); Since $\{v_j = v_\ell, j, \ell = 1, \dots, k\}$ is $[d(N-1)-1]$-rectifiable, and passing from local to global, we conclude.

Rectifiability of Γ. $d^2\mathcal{A}^{\mathrm{ren}}_x[\varphi^x]$ enjoys a *good* spectral theory, with eigenvalues

$$\lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots$$

Points in Γ correspond to points such that $\lambda_1 = 0$ and $\operatorname{Ker}(d^2\mathcal{A}^{\mathrm{ren}}_x) \neq \{0\}$. Thus, we can split Γ in two components $\Gamma' \cup \Gamma''$, defined by

$$\Gamma' = \{x \in \Gamma : \dim(\operatorname{Ker}(d^2\mathcal{A}^{\mathrm{ren}}_x)) = 1\},$$

and Γ'' the subset of Γ in which $\dim(\operatorname{Ker}(d^2\mathcal{A}^{\mathrm{ren}}_x)) \geq 2$.

The idea: for $x_0 \in \Gamma$, define a *local flow* by following time-reversed characteristics, starting from a trasversal section containing $z_0 = \gamma^{x_0}(T_0)$, $T_0 > 1$, where

$$d^2\mathcal{A}^{\mathrm{ren}}_{z_0} \text{ is coercive on } [T_0, \infty) \text{ (hence } \; \lambda_1 > 0).$$

Then define the eigenvalues along this local flow, until reaching a neighborhood of x_0 relative to Γ (thanks to the monotonicity of λ_1).

For Γ', the simplicity of the first eigenvalue, and **Hadamard Formula** allow us to apply again IFT, yelding rectifiability with codimension 1. For Γ'', we can show that $\mathcal{H}^{[d(N-1)-2]^+}(\Gamma'') = 0$, thanks to the *more degeneracy*, and by invoking **Sard Lemma**:

Sard Lemma

Let $F : \mathbb{R}^n \to \mathbb{R}^M$ be a C^∞ map. We have

$$\mathcal{H}^{k+\varepsilon}(F(G_k)) = 0 \quad \text{for every} \quad \varepsilon > 0,$$

where G_k are the points in $\mathbb{R}^n$ such that $\operatorname{rank}(DF) \leq k$.

Size of $\Gamma \setminus \Sigma$ This can be achieved by adapting certain ideas from the time-dependent case in [CS] to our setting, and by suitably reducing the local flow to a transversal direction such that the associated Jacobian matrix retains rank with codimension at least one.

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